1. Briefly describe your original prototype and explain the exact new feature, component, or capability you are adding to it now.

Our Stage 1 project was a VR-Based Immersive Motor Rehabilitation System with Custom Haptic Controller for Physiotherapy. The original prototype consisted of a custom rehabilitation board designed to capture upper-limb movement and interaction, with the overall system intended to integrate movement, grip, applied force and Range of Motion (ROM) with immersive VR-based rehabilitation exercises. The Stage 1 proposal also included therapist monitoring and performance tracking as part of the complete system.

Since our application, we have progressed the project through field testing, design refinement and continued system development. Based on the testing and feedback received, we are currently working on three major developments:

1. Hardware Movement-Range Refinement

The original rehabilitation board was designed for a 0°–180° movement range. During field testing with patients, along with feedback from Dr. Venkata Ramana, Speech Language Pathologist at AIIMS Bhubaneswar, we identified that the required rehabilitation movement should be more controlled around the central position. The hardware is therefore being redesigned for a 60°–120° operating range, corresponding to 30° movement to the left and 30° to the right of the 90° center position.

2. Immersive VR Game Refinement

The immersive rehabilitation game is now being developed according to the refined movement range of the redesigned hardware. We are currently developing the VR rehabilitation environment using Unity and other supporting software, so that the virtual exercises correspond to the updated physical movement requirements of the patient.

3. Rehabilitation Dashboard and Web Platform

We are also developing a dedicated therapist-facing rehabilitation dashboard and complete web-based management platform. The new platform is being developed to organize and visualize rehabilitation information including Range of Motion, Applied Force, Haptic Grip Pressure, Exercise Repetitions, Spatial Position Tracking (X, Y, Z), Motion Status, Patient Records, Therapy Plans, Session Records, Performance Analytics, Hardware Settings, and Sensor/Controller Configuration.

Thus, the current development does not replace our original concept. It refines the physical rehabilitation mechanism based on real-world feedback, synchronizes the VR environment with the refined movement requirement, and adds a digital monitoring and management layer around the rehabilitation system.

2. How does this update improve upon your original work, and what makes your approach smarter, more effective, or more affordable than existing solutions on the market?

The major improvement is that the project is now evolving through actual testing and refinement rather than remaining at the original prototype configuration.

The first major improvement is the hardware movement range. The original 0°–180° design provided a broad movement range, but field testing indicated the need for a more controlled 60°–120° range around the 90° center position. Redesigning the hardware around this requirement allows the physical interaction to be better aligned with the movement requirement identified during testing.

The second improvement is the synchronization between the physical hardware and VR rehabilitation activity. Rather than developing the VR game independently, the new VR environment is being designed around the refined movement range of the hardware. This creates a more consistent relationship between the patient's physical movement and the corresponding virtual rehabilitation activity.

The third improvement is the new rehabilitation dashboard and web platform. Instead of rehabilitation measurements remaining as individual sensor outputs, the platform is being developed to bring ROM, applied force, haptic grip pressure, repetitions, spatial X/Y/Z position and motion status together with patient records, therapy plans, session history and analytics.

This creates a more complete rehabilitation workflow:

Patient → Physical Movement → Custom Hardware → Sensor Data → VR Exercise → Dashboard → Session Records → Analytics → Therapy Management

The approach also remains aligned with the original objective of modular and cost-conscious development. The dashboard is being built around the existing custom rehabilitation hardware and sensing architecture, while the hardware itself is being refined rather than replaced by an entirely different system. The Stage 1 proposal already established a modular hardware/software architecture intended to support additional sensors and rehabilitation functions.

Therefore, the current development can be summarized as:

Original: Custom rehabilitation hardware + sensing + planned VR-based therapy.

Current: Refined rehabilitation hardware + movement-specific VR therapy + real-time rehabilitation dashboard + patient/session management + analytics.

This progression makes the system more controlled, integrated and practical for therapist-guided rehabilitation, while retaining the original hardware-based approach.

3. Based on your lab or field testing, what was your biggest insight or unexpected hurdle, and what immediate next steps are you taking to make the project fully ready for real users?

Our biggest insight from field testing was that the initial hardware movement range needed to be refined according to actual patient movement requirements. The original board was designed for 0°–180°, but testing and feedback from patients and Dr. Venkata Ramana, Speech Language Pathologist at AIIMS Bhubaneswar, highlighted the need to focus the rehabilitation movement around the 90° central position, resulting in the current 60°–120° design requirement.

This insight also affected the software side of the project. Once the physical movement requirement changed, the VR rehabilitation activity could no longer be treated independently; the VR game also needs to be redesigned and developed around the refined movement range.

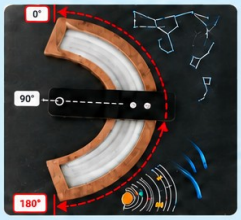
At the same time, as we began working with multiple rehabilitation parameters, another important requirement became clear: collecting data alone is not sufficient. The data needs to be presented and organized in a way that allows therapists to understand patient performance, individual sessions and progress over time. This led to the development of the dedicated rehabilitation dashboard and web platform.

Our immediate next steps are therefore:

1. Complete the redesigned 60°–120° hardware mechanism and refine its mechanical movement.
2. Develop and integrate the VR rehabilitation game according to the refined movement range.
3. Complete the rehabilitation dashboard and web platform, including patient records, therapy plans, session records, analytics and hardware settings.
4. Integrate the hardware sensor data with the dashboard and VR environment.
5. Calibrate and validate ROM, applied force, haptic grip pressure, repetitions and spatial X/Y/Z tracking.
6. Conduct repeated complete-session testing with the refined hardware and software together.
7. Continue therapist-guided validation and refinement before progressing toward real-user deployment.

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1. Original Prototype (Full Board)



- 0° – 180° movement range
- Custom rehabilitation hardware
- Intended VR-based therapy
- Sensor-based data collection (movement, grip, force, ROM)

2. Field Testing (with Patients)



- Tested with patients
- Real-world usage and interaction
- Observed movement requirements
- Collected initial feedback

3. Clinical Feedback

Feedback from Dr. Venkata Ramana (Speech Language Pathologist at AIIMS Bhubaneswar)

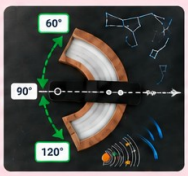
Identified the required movement range as

60° – 120°

(30° left and 30° right of 90°)

4. Current Development (After Clinical Feedback)

4.1 Hardware Movement-Range Refinement



- Redesigned for 60° – 120° range
- 30° left and 30° right of 90°
- More controlled and focused movement based on field testing feedback

4.2 Immersive VR Game Development



- Developing the VR game based on the refined 60° – 120° movement range
- Using Unity and other software
- Virtual exercises synchronized with physical movement

4.3 Rehabilitation Dashboard & Web Platform



- Patient records
- Therapy plans
- Session monitoring
- ROM, Force, Grip Pressure
- Exercise repetitions
- Spatial position (X, Y, Z)
- Motion status
- Performance analytics
- Hardware settings
- Sensor and controller configuration

5. Integration



- Integrate hardware, VR game and dashboard
- Real-time data flow between all components
- Complete rehabilitation workflow

6. Calibration & Testing



- Calibrate sensors and movement range
- Validate ROM, force, grip pressure, repetitions and spatial tracking
- Test complete rehabilitation sessions

7. Therapist Validation



- Further testing with therapists
- Refine based on practical feedback
- Ensure usability and effectiveness

8. Ready for Real Users



- Final refinements
- Fully integrated system
- Prepared for wider deployment